

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2020/2021

MAY 2020 TRIMESTER

FINAL ASSESSMENT

UCCC2063 ALGORITHMS ANALYSIS

THURSDAY, 1 OCTOBER 2020

TIME: 8:30 AM – 11:30 AM (3 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)

Instructions to Students:

REMINDER: You are reminded to read and adhere to the Final Assessment Instructions to Candidates that has been made available through the UTAR Portal before the commencement of Final Assessment (FA). The detailed instructions for this FA are as follows:

General

1. This Final Assessment (FA) is an Individual, time restricted assessment which consists of FOUR (4) questions. Each question carries 25 marks.
2. You are required to answer **ALL** questions, and submit the **ANSWER SCRIPT** by **11.30 am, 1 OCTOBER 2020**.
3. During the period of 3 hours of this FA, the examiner(s) can be reached at
 - (a). [Microsoft Teams] with Code: **rphitzs**
 - (b). Email: **tsesh@utar.edu.my**You may use the above e-platform(s) to check with the examiner(s) if you need any clarification on this FA question paper
4. You may refer to any books, lecture notes, published materials, online resources etc. when answering the questions. Candidates are reminded that **Copy-and-Paste, Consultation, Discussion and Sharing of Answers** with others are **STRICTLY PROHIBITED** in this FA.

UCCC2063 ALGORITHMS ANALYSIS**Answer Script File**

5. The answer script **MUST** be either a Microsoft Word or PDF file, in A4 size format. **Note: Please keep the file size NOT exceeding 5MB.**
6. Please check you index number generated by the Division of Examinations, Awards, and Scholarships (DEAS). You **MUST** name your answer script using the following file name for submission:

[Course Code]_FA_[Programme Abbreviation]_[Your Index Number]

For example, if you are from the degree programme CS, and your Index Number is A01234CBCSF, then your answer script should be named as
UCCC2063_FA_CS_A01234CBCSF.doc

Answer Script Submission

7. Your answer script file has to be submitted to the following platform(s) before the due time/date .
 - (a). Submit your answer script at WBLE.
 - (b). Send your answer script to the course code email:
UCCC2063.FICT@utar.edu.my
Note: For the subject / title of your email, use the file name of your answer script, i.e.,
UCCC2063_FA_CS_[Your Index Number].doc
8. Please make sure you submit the correct, complete and final version of your answer script and the same copy of answer script file is to be submitted to the above platform(s). If different answer scripts are received, the examiner will just randomly choose one of them to mark and the other will be totally ignored.

UCCC2063 ALGORITHMS ANALYSIS

Contents of Answer Script

9. The **first page** of your submission is the **Final Assessment Cover Page**. You **MUST** use the template given and fill in the following information
 - Your Degree Programme (Abbreviation)
 - Your Index Number
 - Your Name
 - Your Student ID
10. The **second page** of your submission is the **Final Assessment Declaration Statement**. You **MUST** use the template given, and digitally sign on the form to indicate the authenticity of your submitted work is without plagiarism.
11. Each question should be answered starting on a new page. It is highly recommended that the answers are placed in same order as the questions.
12. For answer scripts that have text-based answers only, all texts **MUST** be typed and recommended using Times New Roman characters with font size [**e.g. 12**], with proper spacing and alignment, except for the drawings and equations/calculations.
13. For answer scripts that require / contain drawings, equations and calculations with short text descriptions, you can hand-write your answers on a blank paper, and then use the scanner apps in your smartphone (Scanner Document, CamScanner, Adobe Scanner etc.) to take scanned copy, or you can type if necessary and include the scanned copy taken in the Word document, as part of your submitted answers.
14. Please include a page number on **each and every page** of your answer script. Ensure that each page of answer scripts is in sequence prior to online submission.

WARNING OF PLAGIARISM

15. Answer scripts shall be uploaded to Turnitin to check for the originality of submitted answers. In the case of suspected plagiarism, the evidence will be submitted to the Examination Disciplinary Committee of the University. Disciplinary action shall be taken against any candidate who is found to have plagiarized in the answer submitted. Hence, candidates are reminded to abide by all University Rules and Regulations and any instructions/guidelines relating to examinations/assessments.

UCCC2063 ALGORITHMS ANALYSIS**Questions: [Total: 100 Marks]****Q1. [Asymptotic Notations]**

(a) Express the following functions using Big-Oh notation in their best formats.

(i) $2n^{0.5} \log_{30} n^{100} + 22n^{0.7}$ (1 mark)

(ii) $(70n)^{666} + n^{444}(\log_2 n)^{444} + 0.3(2^n)^2 + 2^{5n}$ (1 mark)

(iii) $2(n-1)! + n! + n^7 + n(n!)$ (1 mark)

(iv) $81n^2 + 36n \log_2 n + 2020$ (1 mark)

(v) $33n \log_2 \log_2 \log_2 n + 2n \log_2 \log_2 n$ (1 mark)

(b) Express the following functions using Θ -notation in their best formats.

(i) $2n^{0.5} \log_{30} n^{100} + 22n^{0.7}$ (1 mark)

(ii) $(70n)^{666} + n^{444}(\log_2 n)^{444} + 0.3(2^n)^2 + 2^{5n}$ (1 mark)

(iii) $2(n-1)! + n! + n^7 + n(n!)$ (1 mark)

(iv) $81n^2 + 36n \log_2 n + 2020$ (1 mark)

(v) $33n \log_2 \log_2 \log_2 n + 2n \log_2 \log_2 n$ (1 mark)

(c) Express the following functions using Ω -notation in their best formats.

(i) $2n^{0.5} \log_{30} n^{100} + 22n^{0.7}$ (1 mark)

(ii) $(70n)^{666} + n^{444}(\log_2 n)^{444} + 0.3(2^n)^2 + 2^{5n}$ (1 mark)

(iii) $2(n-1)! + n! + n^7 + n(n!)$ (1 mark)

(iv) $81n^2 + 36n \log_2 n + 2020$ (1 mark)

(v) $33n \log_2 \log_2 \log_2 n + 2n \log_2 \log_2 n$ (1 mark)

(d) Using the definition of Big-Oh, prove $2000n^{100} = O(n^{101})$.[Hint: $O(f(n)) = \{g(n) \mid \exists c, n_0 \bullet \forall n > n_0 \bullet g(n) \leq c f(n)\}$] (5 marks)(e) Using the definition of Ω , prove $2000n^{100} \neq \Omega(n^{101})$.[Hint: $\Omega(f(n)) = \{g(n) \mid \exists c, n_0 \bullet \forall n > n_0 \bullet g(n) \geq c f(n)\}$] (5 marks)

[Total : 25 marks]

UCCC2063 ALGORITHMS ANALYSIS

Q2. [Solving Recurrences]

(a) Using master method to solve the following recurrences and leave your answers in Θ -notations.

(i) $T(n) = 3T(n/4) + n \log n$ (4 marks)

(ii) $T(n) = 4T(n/4) + n \log n$ (4 marks)

(iii) $T(n) = 5T(n/4) + n \log n$ (4 marks)

(b) Using Mathematical Induction (substitution) to prove

(i) $T(n)=O(n)$, for $T(n) = 7T(\sqrt{n}) + n$. (4 marks)

(ii) $T(n)=O(n^3 \log n)$, for $T(n) = 8T(n/2) + n^3$. (4 marks)

(iii) $T(n)=O(n^2)$, for $T(n) = 9T(n/3) + n$. (4 marks)

(c) Can we solve $T(n) = 7T(\sqrt{n}) + n$ by using Master Method? Explain briefly. (1 mark)

[Total : 25 marks]

UCCC2063 ALGORITHMS ANALYSIS

Q3. [Design and Analysis of Algorithm]

Suppose that L1 and L2 are two SORTED linked lists of lengths m and n, respectively.

- (a) Prove that the BEST case time-complexity for merging L1 and L2 into a sorted linked list is $O(1)$. [Hint: give the input scenario, and argue how constant time is enough.] (5 marks)
- (b) Give an $O(m+n)$ worst case time algorithm for merging L1 and L2 into a sorted linked list in pseudocode format, where the time complexity can be easily observed. (10 marks)
- (c) By using the results of (a) and (b), give an algorithm for merging L1 and L2, which worst case time complexity is $O(m+n)$ and the best case time is $O(1)$. [Hint: No matter whether you succeed in working on any of (a) and (b), assume that you have algorithm(s) for them, and apply the algorithm(s).] (5 marks)
- (d) Referring to the algorithm in (c), what is the average case time complexity if the best case scenario occurs with probability 0.999, and the worst case scenario with probability 0.001? Give your answer using Big-Oh notation. [Hint: No matter whether you succeed in working on (c), assume that you have that algorithm, and analyze it.] (5 marks)

[Total : 25 marks]

UCCC2063 ALGORITHMS ANALYSIS

Q4. [Binary Tree]

- (a) Explain how an n-ary tree can be implemented by a binary tree, where integer n is no smaller than three. (3 marks)
- (b) Give the recurrence for the time complexity of the preorder traversal for binary tree. Explain. (No need to solve it.) (3 marks)
- (c) Given a binary search tree of n nodes, of each which contains an integer, what is the optimal time complexity to output the integers in sorted order? Explain why it does not violate the lower bound of worst case time complexity of comparison-based sorting algorithm. (5 marks)
- (d) You are given an array A[1..n], which stores the output of the pre-order traversal of a binary search tree of n integers. In pseudocode format, give an algorithm for constructing the corresponding binary search tree. The pseudocode should illustrate how to manipulate the pointers. You may assume the structure of a node of the binary tree as below:

```
struct node { // for binary tree
    int x;
    node *left, *right;
};
```

[Hint: Use recursive algorithm BuildTree(i, j) to construct the binary tree with preorder traversal stored in A[i...j], and return its root.]

(14 marks)

[Total : 25 marks]

